

TEST CODE **000592**

**FORM TP 9488**

JUNE 1994

**CARIBBEAN EXAMINATIONS COUNCIL**

**SECONDARY EDUCATION CERTIFICATE  
EXAMINATION**

**MATHEMATICS**

**Paper 02 - General Proficiency**

*2 hours 40 minutes*

1. Answer ALL questions in Section I, and any TWO in Section II.
2. Begin the answer for each question on a new page.
3. Full marks may not be awarded unless full working or explanation is shown with the answer.
4. Mathematical tables, formulae and graph paper are provided.
5. Slide rules and silent electronic calculators may be used for this paper.
6. You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.

**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO**

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## SECTION I

Answer ALL the questions in this section.

1. All working must be clearly shown.

- (a) Given that  $a = 4$ ,  $b = -2$ , and  $c = 3$ ,

calculate the value of  $\frac{a^2 - bc}{b + c}$ . ( 2 marks)

- (b) Factorize (i)  $9a^2 - b^2$

(ii)  $3x - 8y - 4xy + 6$ . ( 5 marks)

- (c) Given that  $\frac{2x}{3} + \frac{4}{y} = 1$ , express  $y$  in terms of  $x$ . ( 4 marks)

Total 11 marks

2. (a) Evaluate  $3.7 \times 10^2 + 2.4 \times 10^3$ , giving your answer in standard form. ( 3 marks)

- (b) Janet's gross salary is \$2400 per month. Her tax-free allowances are shown in Table A below.

TABLE A: Tax-free Allowances

|                    |                    |
|--------------------|--------------------|
| National Insurance | 5% of gross salary |
| Personal Allowance | \$3000 per year    |

Calculate

- her gross yearly salary
- her total tax-free allowances for the year
- her taxable yearly income.

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- (iv) A 10% tax is charged on the first \$20 000 of taxable income.  
A 20% tax is charged on the portion of taxable income above \$20 000.

Calculate the amount of income tax Janet pays for a year.

( 8 marks)

Total 11 marks

1. (a) Given the following information:

$$U = \{3, 6, 9, 12, \dots, 27\}$$

$$E = \{\text{even numbers}\}$$

$$G = \{\text{numbers greater than 20}\}$$

$E$  and  $G$  are subsets of  $U$ .

- List the members of  $E$  and of  $G$ .
- Draw a Venn diagram to represent the above data.
- State  $n(E \cup G)$ . ( 6 marks)

- (b) (i) The width of a rectangular field is  $w$  metres. The length is 6 metres more than twice the width.

Write, in terms of  $w$ , algebraic expressions for

- the length of the field
- the area of the field.

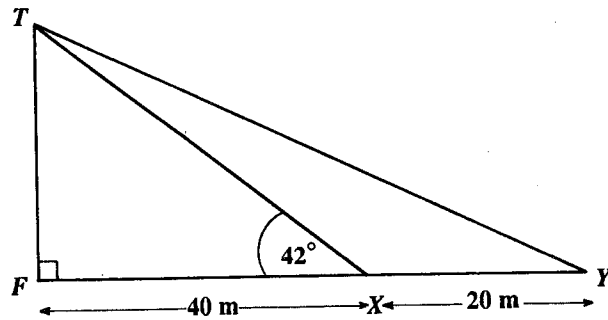
- The area of the field is  $360 \text{ m}^2$ .
  - Write an algebraic equation for the area of the field.
  - Determine the value of  $w$ . ( 6 marks)

Total 12 marks

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4. (a) (i) The scale used for a map is 1 : 250 000. The distance  $MN$  on the map is 4.4 cm.  
Calculate, in kilometres, the actual distance of  $M$  from  $N$ .
- (ii) A car leaves  $M$  at 09:50 hrs and arrives at  $N$  at 10:04 hrs the same day.  
Calculate
- the time, in minutes, taken for the journey
  - the average speed of the car in kilometres per hour, giving your answer to the nearest whole number.
- ( 6 marks)

(b)



In the figure above, **not drawn to scale**,  $TF$  is perpendicular to  $FY$ ,  $FX = 40$  m,  $XY = 20$  m and angle  $TXF = 42^\circ$ .

Calculate, correct to 2 significant figures,

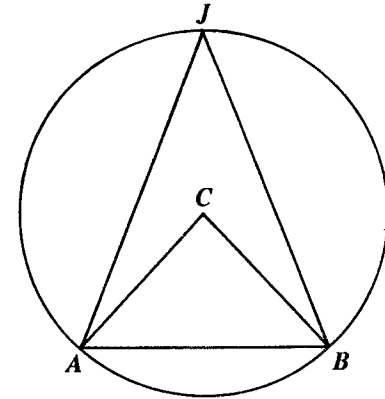
- (i) the length of  $TF$
  - (ii) angle  $XTY$ .
- ( 6 marks)

Total 12 marks

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5. (a) **Using ruler and protractor**, construct a quadrilateral  $VWXY$  in which  $YX = 8$  cm, angle  $XYV = 80^\circ$ ,  $YV = 6$  cm,  $XW = 7$  cm and angle  $XYW = 35^\circ$ .  
Measure and state the length of  $VW$  correct to one decimal place.  
( 5 marks)

(b)



In the figure above, **not drawn to scale**,  $C$  is the centre of the circle of radius 5 cm.  $AB$  is a chord of length 6 cm.  $J$  is a point on the circumference so that  $JA = JB$ .

- (i) Calculate the perpendicular distance of  $C$  from  $AB$ .
  - (ii) Show that  $JA = 9.5$  cm approximately.
- ( 6 marks)

Total 11 marks

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6. (a) The points  $P(2, 1)$ ,  $Q(4, 2)$  and  $R(3, -5)$  are the vertices of  $\Delta PQR$ .  $\Delta PQR$  is mapped onto  $\Delta P'Q'R'$  by a translation  $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ .

(i) On graph paper, using 1 cm to represent 1 unit on each axis, plot the points  $P$ ,  $Q$  and  $R$ .

(ii) Write the position vector of  $P$  in the form  $\begin{pmatrix} a \\ b \end{pmatrix}$ .

(iii) Calculate the vector  $OP'$ , using the sum of two vectors.

(iv) Calculate the coordinates of  $P'$ ,  $Q'$  and  $R'$ .

( 6 marks)

- (b)  $\Delta PQR$  is mapped by a reflection onto  $\Delta P''Q''R''$  where  $P''$  is  $(-4, 1)$  and  $Q''$  is  $(-6, 2)$ .

(i) Plot the points  $P''$  and  $Q''$ .

(ii) Draw on the same axes the mirror line  $\ell$ .

(iii) Write the equation of  $\ell$ .

(iv) Find the coordinates of  $R''$ .

( 5 marks)

Total 11 marks

7. (a) Given that  $\begin{pmatrix} 2 & -3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ y \end{pmatrix} + \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ ,

calculate the values of  $x$  and  $y$ .

( 5 marks)

- (b) A straight line  $HK$  cuts the  $y$ -axis at  $H(0, -1)$ . The gradient of  $HK$  is  $\frac{2}{3}$ .

(i) Show that the equation of the line  $HK$  is  $2x - 3y = 3$ .

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- (ii) On graph paper, using 2 cm to represent 1 unit on each axis, draw on the same axes the graphs of  $2x - 3y = 3$  and  $x + y = 4$

- (iii) Shade only the region to represent both inequalities:

$$2x - 3y \geq 3$$

$$\text{and } x + y \geq 4.$$

- (iv) State the  $x$ -values which satisfy both inequalities.

( 6 marks)

Total 11 marks

8. (a) The results of a test for defective bottles are shown in the table below, where  $x$  is the number of defective bottles per carton, and  $n$  is the number of cartons.

|     |    |    |   |   |   |
|-----|----|----|---|---|---|
| $x$ | 0  | 1  | 2 | 3 | 4 |
| $n$ | 19 | 16 | 5 | 7 | 3 |

Illustrate the information given above on a bar chart.

( 3 marks)

- (b) Each carton contains a total of 48 bottles.

Calculate, for the sample

(i) the total number of bottles

(ii) the total number of defective bottles.

( 4 marks)

- (c) The company produces 250 cartons each day but only ships cartons which contain no defective bottles. Using the results of the above sample,

(i) calculate the probability of randomly choosing a carton which contains no defective bottles

(ii) estimate the number of cartons that is likely to be shipped in a five-day week.

( 4 marks)

Total 11 marks

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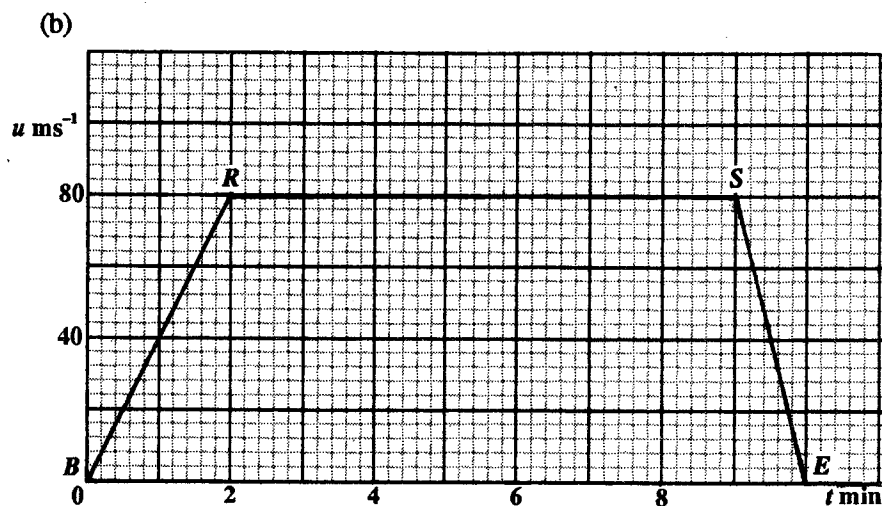
## SECTION II

Answer TWO questions in this section.

## RELATIONS, FUNCTIONS AND GRAPHS

9. Use the answer sheet provided to answer part (a) of this question.

- (a) (i) Indicate, on your answer sheet, the value of  $x$  for which  $2^x = 12$ .  
State this  $x$ -value.
- (ii) Indicate also, the value of  $y$  for which  $y = 2^{1.4}$ .  
State this  $y$ -value.
- (iii) Determine the gradient of the curve  $y = 2^x$  at  $x = 2$ , giving your answer correct to 2 significant figures. (8 marks)



The figure above shows the velocity-time graph  $BRSE$  of a moving body.

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Calculate, clearly stating the units,

- (i) the acceleration of the body during the first two minutes
- (ii) the acceleration of the body during the last minute
- (iii) the total distance moved during the first ten minutes.

( 7 marks)

Total 15 marks

10. (a) Given that  $f(x) = 2x^2 + x - 3$ , copy and complete the table below.

| $x$    | -3 | -2 | -1 | 0 | 1 | 2 |
|--------|----|----|----|---|---|---|
| $f(x)$ | 12 | 3  |    |   |   | 7 |

( 3 marks)

- (b) Using 2 cm to represent 1 unit on the  $x$ -axis, and 1 cm to represent 1 unit on the  $y$ -axis, draw the graph of

$$f(x) = 2x^2 + x - 3 \text{ for } -3 \leq x \leq 2.$$

( 4 marks)

- (c) Using the graph,

- (i) determine the value of  $x$  for the minimum value of  $f(x)$
- (ii) determine the minimum value of  $f(x)$
- (iii) state the values of  $x$  for  $f(x) > 0$ .

( 4 marks)

- (d) Using the same scale and the same axes as in part (b),

- (i) draw the graph of  $g(x) = 3x - 1$
- (ii) shade the area which represents  $f(x) \leq g(x)$ .

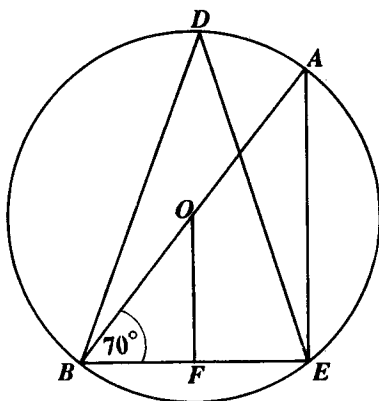
( 4 marks)

Total 15 marks

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## TRIGONOMETRY AND GEOMETRY

11. (a)



In the diagram above, not drawn to scale,  $O$  is the centre of the circle and  $AOB$  is a diameter.  $D$  is a point on the circumference and  $F$  is the midpoint of the chord  $BE$ . Angle  $ABE = 70^\circ$ .

- Calculate angle  $BDE$ .
- Show that  $\triangle OFB$  and  $\triangle AEB$  are similar.
- Calculate angle  $AOF$ .

(Note: State reasons and show necessary working.)

( 9 marks)

(b)  $VMNPQ$  is a pyramid on a square base  $MNPQ$  of side 40 cm.

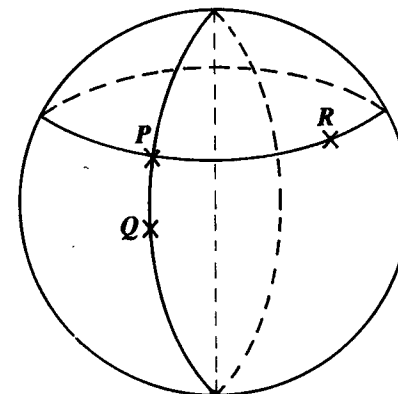
- Draw a diagram to represent the pyramid. Clearly label the vertices.
- Draw a plan of the pyramid, viewed from above. State the scale used.
- The height of the pyramid is 20 cm. Show that the length of the sloping edge  $VM$  is  $20\sqrt{3}$  cm.

( 6 marks)

Total 15 marks

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12. (a)



(In this question take the radius of the earth to be 6400 km and  $\pi$  to be 3.142.)

- A ship  $X$  travelled due South from  $P$  ( $15^\circ$  W,  $30^\circ$  N) to  $Q$ .
  - Calculate the circumference of the circle that passes through  $P$  and  $Q$ .
  - Given that the distance  $PQ$  is 3114 km, calculate the position of  $Q$ .
- Another ship  $Y$  travelled due East from  $P$  ( $15^\circ$  W,  $30^\circ$  N) to  $T$  ( $30^\circ$  E,  $30^\circ$  N).
  - Show that the radius of the circle that passes through  $P$  and  $T$  is  $6400 \cos 30^\circ$ .
  - Calculate the distance  $PT$ .

( 7 marks)

(b) In triangle  $VXR$ ,  $VX = 5$  cm,  $VR = 4$  cm and angle  $XVR = 55^\circ$ . Calculate to 3 significant figures

- the area of triangle  $VXR$
- the perpendicular distance of  $X$  from  $VR$
- the magnitude of angle  $XRV$ .

( 8 marks)

Total 15 marks

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## VECTORS AND MATRICES

13.  $OFGH$  is a parallelogram where  $O(0, 0)$  is the origin. The position vector of  $F$  is  $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ , and the position vector of  $G$  is  $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ .  $M$  is the midpoint of  $OG$ .

- (a) Sketch a diagram to show the position vectors of  $F$  and of  $G$ .  
( 2 marks)
- (b) Determine the vector  $\overrightarrow{FG}$ .  
( 3 marks)
- (c) Show that the position vector  
(i) of  $H$  is  $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$   
(ii) of  $M$  is  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ .  
( 2 marks)
- (d) Use vectors to show that  
(i) the diagonals of the parallelogram  $OFGH$  bisect one another  
(ii)  $OFGH$  is a square.  
( 8 marks)

Total 15 marks

14. (a) Show, with the aid of diagrams, that

- (i) the transformation represented by the matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  is NOT a rotation.  
(ii) the transformation represented by the matrix  $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$  is NOT a reflection.

( 4 marks)

- (b) Determine the matrix which represents a reflection in the line  $y = -x$ .  
( 5 marks)

- (c) The matrix  $A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$  and the matrix  $K = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ .

- (i) Calculate  $AK$ .  
(ii) Identify the coordinates of the image of the point  $(5, 3)$  under the combined transformation represented by  $AK$ .  
(iii) Describe completely the combination of the two transformations represented by  $AK$ .  
( 6 marks)

Total 15 marks

END OF TEST

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Paper 02 - General Proficiency

Answer Sheet for Question 9 (a)

Candidate Number .....

